

# Phuong Huu Vu Tran

Undergraduate Researcher • Multimodal & Reliable AI

10425032@student.vgu.edu.vn • Ho Chi Minh City, Vietnam

Google Scholar • ORCID • GitHub • LinkedIn

## RESEARCH PROFILE

Second-year Computer Science undergraduate at Vietnamese-German University, with research experience at PAMI Lab and RAISE Lab. Research focuses on multimodal and vision-language learning, reliable AI, evidence verification, and uncertainty-aware decision making, with work spanning multimodal retrieval, robust and reliable RAG, claim decomposition, argument mining, and video prediction. Interested in extending these methods to healthcare AI.

## EDUCATION

**Vietnamese-German University**

*B.Sc. in Computer Science — Foundation Year GPA: 9.0/10*

2025–2029 (expected)

Ho Chi Minh City, Vietnam

**Le Quy Don High School for the Gifted**

*Specialization in Informatics*

2022–2025

Khanh Hoa, Vietnam

## RESEARCH EXPERIENCE

**Undergraduate Research Assistant — PAMI Lab**

*Vietnamese-German University*

Sep 2025–Present

Supervisor: Dr. Cuong Tuan Nguyen

- Co-developed FaLCon and led its Qwen3-based reranking, uncertainty gating, and late-consensus inference; the system achieved 95.41% mAP@10 on PAB.
- Contributed pipeline design and selected experiments for GeoRoute, a geometry-aware traffic future-frame prediction system.

**Visiting Undergraduate Research Assistant — RAISE Lab**

*Ho Chi Minh City University of Technology*

Jan 2026–Present

Supervisor: Dr. Xuan-Bach Le

- Co-authored LTM-RAG, a longitudinal trust-mining framework for poison-resilient RAG, accepted to the Research Track of IEEE ICDM 2026.
- Developed a risk-controlled evidence verification framework for noisy and updatable RAG using con-formal Answer/Fallback/Abstain decisions.
- Developed CREDENCE, including the Semantic-F1 metric and a verified claim-decomposition and repair framework with formal convergence analysis and cross-domain evaluation.

## PUBLICATIONS

- [P1] Thi-Hong-Cuc Le, **Phuong Huu Vu Tran**, Thuan Mai Duc, and Xuan-Bach Le.  
*LTM-RAG: Longitudinal Trust Mining for Poison-Resilient Retrieval-Augmented Generation.*  
IEEE International Conference on Data Mining (ICDM 2026), Research Track **Second author**  
(accepted).
- [P2] **Phuong Huu Vu Tran**, Long Minh Vo, Son Nguyen Minh Le, and Hoang Van.  
*LLM-INSTRUCT at UZH Shared Task 2026: Constraint-Aware Retrieval and Selective Debate for Paragraph-Level Argument Mining.*  
Proceedings of the 13th Workshop on Argument Mining and Reasoning **First author**  
(ArgMining 2026), co-located with ACL 2026. [Paper]
- [P3] Hieu Dinh Trung Pham\*, **Phuong Huu Vu Tran\***, Thuan Duc Mai, Son Nguyen Minh Le, Khang Le Minh, Hoang Vo, Minh-Chi Phung, Huy Minh Nhat Nguyen, and Cuong Tuan Nguyen.  
*FaLCon: Facet-Anchored Retrieval with Late Consensus for Sim2Real Text-Based Person Anomaly Search.*  
AI City Challenge Workshop, ECCV 2026 (accepted). [arXiv] **Co-first author**

- [P4] Khang Minh Le, Hieu Dinh Trung Pham, Luu Thanh Danh, Nam-Tien Le, Hieu Anh Ngo, **Phuong Huu Vu Tran**, Son Nguyen Minh Le, Nguyen Trong Nghia, Tu Tran Thi Cam, Huy Minh Nhat Nguyen, and Cuong Tuan Nguyen.  
*GeoRoute: Geometry-Aware Hybrid Inference for Traffic Future-Frame Prediction*.  
 AI City Challenge Workshop, ECCV 2026 (accepted). [arXiv] Co-author

\*Equal contribution.

## MANUSCRIPTS UNDER REVIEW

---

- [M1] *Risk-Controlled Evidence-Regime Verification for Noisy, Updatable RAG Fact Checking: A Benchmark and a Conformal Answer/Fallback/Abstain Protocol*.  
 Journal of Artificial Intelligence Research (JAIR). First author • Under review
- [M2] *CREDENCE: Claim Reduction for Decomposition & Enhanced Credibility – Semantic Metrics and Convergence Analysis*.  
 Language Resources and Evaluation (LRE). [arXiv] First author • Under review
- [M3] *SCoRe: Smoothed Certified Release for Prompt Repair*.  
 Neural Information Processing Systems (NeurIPS 2026). Co-first author • Under review

## RESEARCH & DEVELOPMENT EXPERIENCE

---

**AI Research & Development Intern — Transformers Lab** Jan 2026–Present  
 VBI Academy Supervisor: Assoc. Prof. Dr. Ngoc-Thanh Dinh

- Contributed to the development of the xfmr-zem pipeline, a legal-document RAG system, and an end-to-end data-processing pipeline for corpus construction.
- Conduct ongoing research on neuro-symbolic AI at Transformers Lab.

## SELECTED AWARDS & HONORS

---

- Champion (1st Place)** — UZH Shared Task, ArgMining Workshop 2026
- Promising Award (Semifinalist)** — VTV AI Thuc Chien 2026
- Bronze Medal** — Vietnam Olympiad of Artificial Intelligence (VOAI) 2025
- Silver Medal** — AI Challenge, 6th Central Vietnam–Central Highlands Informatics Olympiad 2025
- First Prize** — Khanh Hoa Provincial Science and Engineering Fair 2024

## TECHNICAL SKILLS

---

- Programming & ML:** Python, C++, PyTorch
- Multimodal & Vision:** vision-language models, CLIP-based retrieval, video prediction, diffusion models, depth-based rendering
- LLMs & Reliable AI:** LLM reranking, retrieval-augmented generation, evidence verification, uncertainty calibration, conformal prediction
- Tools:** Git, Linux, LaTeX/XeLaTeX

## LANGUAGES

---

**Vietnamese:** Native • **English:** IELTS Academic 6.5 • **German:** B1 (self-assessed)